

PENG, PEI (彭沛)

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Date of Birth: 24 November 1998



Research Summary

I am a second-year Ph.D. student mentored by Prof. Sheng-Jun Huang in the Weakly Supervised Learning Laboratory at Nanjing University of Aeronautics and Astronautics (NCAA). My research uses ideas from **philosophy thought** (causality, counterfactual reasoning) in **multimodal model** to improve their interpretability and reliability, and to **address the problem of hallucinations** where models was misled into producing fake results by its inherent biases.

Education

Ph.D. —College of Computer Science and Technology (NCAA) 2024.4 – 2028.4 (expected)

- Major: Computer Science and Technology
- Research fields: By introducing **philosophy concepts** such as **causal relationships**, it can make **multimodal model** more reliable (robust) and explainable (transparent), particularly interested in model hallucinations and model shortcut learning.

Research Assistant —Weakly Supervised Learning Laboratory (NCAA) 2023.4 – 2024.4

- Research fields: Applications of causality in multi-label domains, such as label co-occurrence and label noise

Master —College of Energy and Power Engineering (NCAA) 2020.9 – 2023.4

- Major: Aeronautical Science and Technology (Systems Engineering)
- Research fields: Non-destructive analysis of air systems (e.g., aircraft) to assess fault sources and lifecycle of machine systems.

Bachelor —College of Energy and Power Engineering (NCAA) 2016.9 – 2020.6

- Major: Energy and Power Engineering

Publications

- Peng Pei, Xie MingKun, et al. Representation-Level Counterfactual Calibration for Debiased Zero-Shot Recognition [C]. Advances in Neural Information Processing Systems, 2025.
- Xie MingKun, Peng Pei, et al. Counterfactual reasoning for multi-label image classification via patching-based training[C]//Proceedings of the 41st International Conference on Machine Learning. 2024.
- Peng Pei, Zhao YongPing. Robust semi-supervised discriminant embedding method with soft label in kernel space[J]. Computing and Applications, 2022: 1-23.

Experience and Awards

- Passed College English Test Band 6 (CET-6)、 IELTS Score: 6.5
- Participant, IJCAI Youth Exchange Forum, Shanghai, 2023.
- Participant, Jiangsu Computer Society Meeting, Liyang, 2022.
- Second Prize (University Level), National College Student Mathematical Modeling Competition, 2020.
- Reviewer for NeurIPS-2025; NeurIPS-2024; ACM MM-2025; ICME-2025; ECCV-2024; CVPR2025